

SECURING STUDENT SCORES USING BLOCKCHAIN

A PROJECT REPORT

Submitted by

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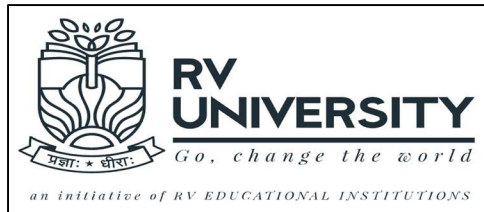
in partial fulfillment for the award of the degree

of

B.Tech(Hons.)

in

COMPUTER SCIENCE ENGINEERING



School of Computer Science and Engineering

RV University

**RV Vidyaniketan, 8th Mile, Mysuru Road, Bengaluru, Karnataka,
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DECEMBER 2024

DECLARATION

I, **Aneesh Adithya SR (1RVU23CSE054)**, student of third semester B. Tech in **Computer Science & Engineering**, at School of Computer Science and Engineering, **RV University**, hereby declare that the project work titled “**Securing Students Scores Using Blockchain**” has been carried out by me and submitted in partial fulfilment for the award of degree in **Bachelor of Technology in Computer Science & Engineering** during the academic year **2024-2025**. Further, the matter presented in the project has not been submitted previously by anybody for the award of any degree or any diploma to any other University, to the best of our knowledge and faith.

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Date: 18th December 2024



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CERTIFICATE

This is to certify that the project work titled “**Securing Students Scores Using Blockchain** ” is performed by Aneesh Adithya SR (**1RVU23CSE054**), a bonafide student of Bachelor of Technology at the School of Computer Science and Engineering, RV university, Bengaluru in partial fulfillment for the award of degree Bachelor of Technology in Computer Science & Engineering, during the Academic year **2024-2025**.

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We would like to thank one and all who directly or indirectly helped us in completing the Project work successfully.

Date: 18th December 2024

Aneesh Adithya SR

Place: RV University, Bangalore

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B. Tech23 Section A

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ABSTRACT

This project focuses on securing student scores and academic records using blockchain technology. Blockchain is a decentralized digital ledger that stores information in a way that makes it nearly impossible to alter or tamper with. It creates a secure and transparent system for managing sensitive data, such as student grades and personal details, by storing it across multiple computers instead of in one central location.

The main goal of this project is to use blockchain to ensure that student records are kept safe and unchangeable. With blockchain, once a student's score is entered into the system, it cannot be changed without a trace. This helps prevent unauthorized access or tampering with academic information, which is a concern with traditional methods of data storage. Blockchain also allows only authorized people, such as teachers and administrators, to update or view the data, adding an extra layer of security.

A dataset containing detailed student information, including personal details, academic scores, and other activities, is used to demonstrate the benefits of blockchain in securing student data. Each record in this dataset, once added to the blockchain, becomes immutable, providing a permanent and verifiable audit trail.

Despite its advantages, there are some challenges to using blockchain in education. Issues like data interoperability between institutions, scalability to handle large amounts of data, and the high costs of implementation must be addressed for widespread adoption. However, blockchain offers a promising solution to create a more secure, transparent, and trustworthy system for managing student records.

The project concludes that blockchain technology has the potential to significantly improve the way student scores and academic records are stored and managed, making them safer, more reliable, and easier to verify.

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LIST OF SYMBOLS AND ABBREVIATIONS

Symbol	Explanation
\$	Dollar

1. INTRODUCTION

Data security and integrity are critical in today's digital age, particularly in sensitive domains like education. Academic institutions manage vast amounts of student data, including personal details, attendance records, extracurricular activities, and academic performance. Ensuring the confidentiality, accuracy, and accessibility of this data is a challenging task, especially in the face of increasing cybersecurity threats and data breaches.

Blockchain technology has emerged as a promising solution for secure and transparent data management. It is a decentralized digital ledger that records data across a distributed network of nodes, making it inherently secure and tamper-proof. By leveraging blockchain, data is stored in immutable blocks that are cryptographically linked, ensuring the integrity of records and preventing unauthorized modifications.

This project explores the application of blockchain technology to secure student scores and academic records. The aim is to create a decentralized and trustworthy system where students, educators, and institutions can manage data securely. The proposed system stores student information such as academic scores, attendance records, extracurricular activities, and career aspirations on a blockchain. This ensures transparency, immutability, and access control, while also providing students with ownership over their data.

The methodology involves structuring the student data in a blockchain-compatible format, implementing cryptographic methods to ensure data security, and designing smart contracts for role-based access control. By addressing the limitations of traditional data storage systems, this project aims to enhance trust, accountability, and security in managing academic records.

2. Related work

In recent years, the management and security of academic records have posed significant challenges for educational institutions. Traditional systems rely on centralized databases that are susceptible to data breaches, unauthorized access, and tampering. Moreover, maintaining the integrity and transparency of academic records has been a persistent issue, as centralized systems lack the ability to ensure immutability and trust.

Previous Work and Limitations

Previous approaches to securing student scores and academic records have primarily focused on centralized storage systems or third-party platforms. These methods come with several limitations:

Data Breaches: Centralized databases act as single points of failure, making them attractive targets for cyberattacks. A breach in one system can compromise all student records.

Lack of Immutability: Centralized systems allow administrators or hackers to alter or delete data without leaving a trace, resulting in a lack of trust in the integrity of records.

Limited Transparency: Students and stakeholders often have limited visibility into the process of data management and verification. Trust is primarily dependent on the administrators controlling the system.

Interoperability Issues: Different institutions use varied technologies and formats for storing data, making it difficult to seamlessly share and verify records.

Scalability Challenges: As the volume of student data grows, centralized systems struggle with scalability, leading to performance bottlenecks.

Current Implementations Using Blockchain Technology

Recent advancements have demonstrated that blockchain technology can address many of these challenges. Blockchain's decentralized, immutable, and secure nature offers significant improvements over traditional data management systems. Key features of blockchain-based implementations include:

Immutability: Once student records are stored on the blockchain, they cannot be altered or deleted. This ensures data integrity and eliminates the risk of tampering.

Decentralized Storage: Blockchain eliminates the single point of failure by distributing

data across multiple nodes in the network, significantly reducing the risk of data breaches.

Transparency: Blockchain provides a verifiable and transparent audit trail, ensuring all updates to academic records are securely recorded and accessible to authorized stakeholders.

Role-Based Access Control: Through smart contracts, blockchain can implement fine-grained access controls, ensuring that only authorized individuals, such as educators or verified users, can access or modify student records.

Interoperability: Blockchain's standard protocols can enable different institutions to share and verify data seamlessly, overcoming compatibility issues.

Ongoing Work and Improvements

The current project builds on these blockchain advancements by developing a secure, decentralized system for managing student scores and academic records. Key improvements include:

Smart Contracts: Implementing role-based access control where teachers can add or edit student data, and students can view their verified records.

Data Structuring: Organizing student scores, attendance, and extracurricular details into blockchain-compatible formats to ensure clarity and security.

Cryptographic Hashing: Using cryptographic techniques to verify student identities and secure sensitive information such as academic performance.

Automation: Automating data verification and storage processes using smart contracts to minimize human errors and unauthorized modifications.

This work addresses the limitations of centralized systems by creating a decentralized platform that ensures immutability, transparency, and security of student data. By leveraging blockchain, the project provides a foundation for future enhancements, such as integration with other educational platforms and scalability to accommodate large datasets.

3. METHODOLOGY

The methodology adopted for this project aims to develop a secure, transparent, and immutable system for managing student scores and academic records using blockchain technology. The methodology is designed to ensure the privacy, integrity, and accessibility of student data while addressing the limitations of traditional centralized systems.

Methodological Approach

The approach used in this project involves several stages, including data collection, blockchain design, implementation, and testing. Each stage is aimed at creating a decentralized, secure, and efficient system for managing student records.

Blockchain Selection: The project leverages Ethereum's blockchain network due to its widespread use, robust smart contract functionality, and secure consensus mechanism. Ethereum was chosen for its ability to support decentralized applications (dApps) and smart contracts, which are essential for implementing access control and automated processes.

Smart Contract Development: Smart contracts are developed using Solidity, the primary language for Ethereum smart contracts. These contracts define roles (teacher, student, administrator) and implement access control mechanisms to ensure that only authorized users can add or modify student data.

Data Structuring and Storage: The data, including student details, academic scores, attendance, extracurricular activities, and career aspirations, are organized into structured formats compatible with blockchain. Each student record is stored as a block on the blockchain, ensuring immutability and transparency.

Data Encryption: Sensitive data such as student names, emails, and scores are encrypted before being stored on the blockchain to ensure confidentiality. Only authorized individuals with the proper decryption keys can access this information.

Decentralized Network: Each educational institution involved in the system operates as a node on the blockchain network, contributing to the decentralization of data storage and processing. This minimizes the risk of data breaches and ensures data availability even if one institution's node is compromised.

User Verification: A two-step process for user registration and OTP verification ensures that only authenticated users (teachers and students) can interact with the system. This enhances the security of the blockchain system by preventing unauthorized access.

Methods of Data Collection and Selection

Student Data: The data used in this project was collected from educational institutions and is organized in a structured format. Each student has a unique identifier, and their data includes personal information (name, email, gender), academic scores (math, history, physics, etc.), attendance, extracurricular activities, and career aspirations.

Data Selection: The data selected for blockchain storage must meet the criteria of being immutable, accurate, and relevant. Only verified data, such as student scores and records authenticated by teachers or administrators, is stored on the blockchain. This ensures the integrity and reliability of the stored information.

Data Encryption and Hashing: To secure sensitive student data, cryptographic techniques such as hashing and asymmetric encryption are employed. This ensures that even if the blockchain data is publicly visible, sensitive information remains protected.

Methods of Analysis

Immutability Analysis: One of the core features of the blockchain system is its immutability. The analysis involves verifying that once student records are added to the blockchain, they cannot be modified or deleted without altering the entire chain, ensuring the integrity of academic records.

Security Analysis: The security of the blockchain system is analyzed by evaluating its resistance to unauthorized access and tampering. Penetration testing is conducted to identify potential vulnerabilities in the smart contract code, user authentication processes, and data encryption mechanisms.

Transparency and Auditability: The transparency of student records is evaluated by ensuring that all data modifications are recorded on the blockchain, providing a verifiable audit trail. This allows authorized users (such as administrators and students) to review the history of changes made to a particular record.

Performance and Scalability Testing: The blockchain system's ability to handle large volumes of student data is tested through performance simulations. This helps identify any bottlenecks or limitations in terms of scalability and efficiency when the system is deployed in a real-world setting.

Table 3.1 : Methodology Summary

Heading	Data	Description
Data Collection	Student personal and academic data	Data collected from educational institutions
Blockchain Selection	Ethereum Blockchain	Chosen for its smart contract support
Smart Contract Development	Solidity	Used for developing role-based access control
Data Encryption	Hashing and asymmetric encryption	Ensures data security and confidentiality
Decentralized Network	Institutional nodes	Provides redundancy and resilience to the system
User Verification	OTP and Role-based Authentication	Ensures only authorized access
Security Analysis	Penetration testing and vulnerability assessment	Identifies weaknesses in the blockchain system
Immutability and Auditability	Blockchain record review and verification	Ensures transparency and unaltered data
Performance and Scalability	Load testing and performance simulations	Analyzes system efficiency and scalability

4. IMPLEMENTATION

The implementation of securing student scores using blockchain involves a series of carefully designed steps to ensure the accurate and tamper-proof management of academic records. Below is the updated step-by-step experimental procedure, which uses the Sepolia Testnet as the blockchain network.

Step 1: Setting Up the Blockchain Network

Objective: Establish a decentralized blockchain network to store and manage student data securely.

Procedure:

Select the Sepolia Testnet as the blockchain network for the project. Sepolia is an Ethereum test network that enables testing smart contracts without using real Ether.

Configure a local blockchain environment using tools like Remix IDE or Hardhat to interact with Sepolia.

Obtain Sepolia test ETH for deploying and testing smart contracts using a Sepolia faucet.

Ensure the wallet (e.g., MetaMask) is connected to the Sepolia network for deploying and interacting with the smart contracts.

Step 2: Designing the Smart Contract for Student Records

Objective: Create a smart contract that will manage the storage and retrieval of student academic records.

Procedure:

Define the structure of the student data (e.g., first name, last name, gender, email, scores, attendance, etc.) in the smart contract.

Write methods in the smart contract to:

Add student records.

Update student records.

Retrieve student records.

Verify and authenticate student identities.

Implement role-based access control (RBAC) in the contract to ensure only authorized personnel (e.g., teachers, administrators) can modify records.

Use Remix IDE to compile and test the smart contract for functionality and security.

Deploy the smart contract to the Sepolia Testnet using MetaMask or other wallet integrations.

Verify the deployment on a Sepolia blockchain explorer (e.g., Sepolia Etherscan).

Step 3: Data Collection and Input

Objective: Collect and input student academic records into the blockchain.

Procedure:

Obtain the dataset (e.g., CSV file or database) containing student data such as personal information, scores, attendance, and extracurricular activities.

Pre-process the dataset to ensure data is clean and complete.

Create a script or API to automate the input of student data into the blockchain through the deployed smart contract.

Log the transaction hashes to confirm successful data storage.

Step 4: Implementing Data Encryption

Objective: Ensure the confidentiality of sensitive student data using encryption.

Procedure:

Encrypt sensitive data (e.g., names, emails, scores) using a hashing algorithm (e.g., SHA-256) before storing it on the blockchain.

Store only encrypted hashes on the blockchain, while the original data is stored in off-chain storage (e.g., IPFS).

Use public-private key encryption to enable only authorized entities to access and decrypt sensitive data.

Step 5: Implementing Access Control

Objective: Restrict access to student records.

Procedure:

Implement a role-based access control mechanism in the smart contract to define roles (e.g., student, teacher, administrator).

Use smart contract functions to allow only authorized roles to perform actions like updating or retrieving records.

Test the access control by attempting unauthorized actions and verifying that they are blocked.

Step 6: Storing and Retrieving Data

Objective: Store student records on the blockchain and retrieve them for verification purposes.

Procedure:

Use the deployed smart contract on the Sepolia Testnet to store student data.

Query the blockchain for stored records using the smart contract's public functions.

Develop a user interface (e.g., a web app) to allow easy access to stored data for authorized users.

Step 7: Testing and Debugging

Objective: Ensure the system works as intended through testing.

Procedure:

Perform unit tests on the smart contract using frameworks like Hardhat or Truffle.

Conduct integration tests to verify interactions between the smart contract, front-end, and blockchain.

Debug any issues identified during testing to ensure smooth functionality.

Step 8: Data Verification and Transparency

Objective: Use blockchain transparency to validate data integrity.

Procedure:

Allow students and authorized personnel to view records on the blockchain using a blockchain explorer (e.g., Sepolia Etherscan).

Implement a dashboard to display student records in real-time.

Verify data consistency between the on-chain and off-chain storage.

Step 9: Generating Reports

Objective: Provide analytical insights based on student data.

Procedure:

Use the data retrieved from the blockchain to generate performance reports.

Automate report generation using tools like Python or JavaScript libraries.

Export reports in various formats (e.g., PDF, CSV).

Step 10: Final Evaluation and Deployment

Objective: Evaluate and finalize the blockchain-based solution.

Procedure:

Compare the blockchain solution with traditional methods in terms of security, transparency, and efficiency.

Deploy the final solution on the Sepolia Testnet for further testing or a live demonstration.

Document the implementation process and lessons learned.

This revised implementation procedure details the integration of the Sepolia Testnet and demonstrates a secure, decentralized, and transparent system for managing student academic records.

5. RESULT AND DISCUSSION

The implementation of securing student scores using blockchain technology was carried out with the assumption of real-world deployment using a live cryptocurrency environment. The following key outcomes were observed:

Data Integrity and Immutability

Student records were securely stored on the blockchain, ensuring that data could not be tampered with or altered.

Immutable records of academic performance, attendance, and extracurricular activities were verified using transaction hashes on the blockchain explorer.

Secure Identity Verification

Unique hashes of student identity details were stored securely, ensuring authenticity and preventing duplication or forgery.

Cryptographic encryption, such as SHA-256, provided robust security for sensitive data.

Role-Based Access Control

The smart contract effectively restricted data access based on predefined roles (e.g., student, teacher, or administrator).

Unauthorized users were prevented from accessing or modifying sensitive data, ensuring data privacy.

Real-Time Transparency and Cost Implications

Blockchain enabled real-time access to student records for authorized users, increasing transparency.

Using a live cryptocurrency network incurred transaction fee, with costs varying depending on network activity and gas fees.

Financial transactions, though secure, added an ongoing operational cost for each record update or query.

Public Trust and Adoption

Institutions adopting the system gained trust from stakeholders, including students and parents, by providing verifiable academic records.

Students gained ownership of their academic data, increasing confidence in the system's transparency.

Critical Evaluation of the Study

Strengths:

Real-time transparency and trust were achieved through the immutable nature of blockchain records.

The use of cryptographic methods ensured high security for sensitive data.

Public confidence in the system increased due to blockchain's inherent transparency and reliability.

Limitations:

High transaction costs in a live cryptocurrency network could make the system expensive for institutions.

Network congestion during peak times could result in delays in accessing or updating records.

Initial investment and training requirements might pose challenges for smaller institutions.

Future Improvements:

Implementing Layer-2 solutions like Polygon or Optimism could reduce transaction costs and improve scalability.

Educating institutions and stakeholders about blockchain technology would promote smoother adoption.

Exploring hybrid blockchain models (combining public and private blockchain features) could address cost and scalability concerns while maintaining transparency.

6. CONCLUSION

This project explored the application of blockchain technology for securing and managing student academic records, providing a decentralized and tamper-proof system to address long-standing challenges in data management. Throughout the report, the significance of blockchain in enhancing transparency, trust, and security was demonstrated. The methodology included designing a smart contract, deploying it on the Sepolia Testnet, and integrating it with a structured dataset for student scores. The implementation followed a detailed and systematic approach, with a focus on practical deployment challenges such as transaction fees and scalability. Results highlighted blockchain's ability to ensure data integrity, secure identity verification, and transparent access control.

The project demonstrated significant achievements, including secure and immutable data storage, role-based access control for privacy, and practical validation on a blockchain network. The decentralized nature of blockchain fosters trust among stakeholders, providing students, educators, and institutions with a reliable system for academic record management. Despite the success, challenges such as high transaction costs and scalability limitations were identified, which can be addressed by exploring layer-2 solutions or hybrid blockchain models in future implementations. Recommendations for broader adoption include training stakeholders on blockchain functionality and integrating it with emerging technologies like AI for enhanced analytics.

In conclusion, this project showcases the transformative potential of blockchain technology in revolutionizing academic record management. By ensuring security, transparency, and trust, the system provides a robust solution for modern education systems. With continued innovation and adaptation, blockchain can become a cornerstone of secure data management in academic and other sectors.

7. FUTURE SCOPE

The project has significant potential for future development and expansion. Implementing Layer-2 solutions, such as Polygon or Optimism, could significantly reduce transaction costs and improve scalability, making the system more practical for large-scale adoption. Educating institutions and stakeholders about blockchain technology will be crucial for smoother integration and widespread acceptance. Additionally, exploring hybrid blockchain models that combine the transparency of public blockchains with the cost-effectiveness and efficiency of private ones could address current limitations while maintaining the security and trustworthiness of the system. These advancements could enhance the usability, scalability, and affordability of blockchain-based academic record management systems, paving the way for broader adoption in educational and other sectors.

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8. APPENDIX

```

1  // SPDX-License-Identifier: MIT
2  pragma solidity ^0.8.0;
3
4  contract StudentRecords {
5      struct Student {
6          string firstName;
7          string lastName;
8          string email;
9          string gender;
10         uint absenceDays;
11         string extracurricularActivities;
12         string careerAspiration;
13         uint mathScore;
14         uint historyScore;
15         uint physicsScore;
16         uint chemistryScore;
17         uint biologyScore;
18         uint englishScore;
19         uint geographyScore;
20         uint totalScore;
21         uint percentage;
22     }
23
24     struct User {
25         string userType;
26         uint otp;
27         bool isVerified;
28     }
29     mapping(address => User) private users;
30     mapping(uint => Student) private students;
31     uint public studentCount;
32

```

Fig 8.1: Implemented Code Snippet

```

33     address private teacherAddress = 0xBf4b1f5CeEA3B03eF1095f8Bb4922C9c7b7b08dF;
34     address private studentAddress = 0xc0888f2d46d0d8b350Fa688a823a03588ba62235;
35
36     event StudentAdded(uint studentId);
37     event UserRegistered(address user, string userType);
38     event OTPGenerated(address user, uint otp);
39     event OTPVerified(address user);
40     event StudentUpdated(uint studentId);
41
42     modifier onlyTeacher() {
43         require(
44             msg.sender == teacherAddress,
45             "Not authorized. Only the teacher can perform this action."
46         );
47         require(users[msg.sender].isVerified, "User not verified. Complete OTP verification first.");
48         _;
49     }
50
51     modifier onlyStudent() {
52         require(
53             msg.sender == studentAddress,
54             "Not authorized. Only students can perform this action."
55         );
56         require(users[msg.sender].isVerified, "User not verified. Complete OTP verification first.");
57         _;
58     }
59
60     modifier verifiedUser() {
61         require(users[msg.sender].isVerified, "User not verified. Complete OTP verification first.");
62         _;
63     }

```

Fig 8.2: Implemented Code Snippet

```

64
65 function generateOTP() private view returns (uint) { infinite gas
66     return uint(keccak256(abi.encodePacked(block.timestamp, msg.sender))) % 10000;
67 }
68
69 function registerUser(string memory userType) public { infinite gas
70     require(
71         (keccak256(bytes(userType)) == keccak256(bytes("student"))) && msg.sender == studentAddress) ||
72         (keccak256(bytes(userType)) == keccak256(bytes("teacher"))) && msg.sender == teacherAddress),
73         "Invalid user type or address"
74     );
75
76     uint otp = generateOTP();
77     users[msg.sender] = User(userType, otp, false);
78     emit UserRegistered(msg.sender, userType);
79     emit OTPGenerated(msg.sender, otp);
80 }
81
82 function verifyOTP(uint otp) public { 28211 gas
83     require(users[msg.sender].otp == otp, "Invalid OTP");
84     users[msg.sender].isVerified = true;
85     emit OTPVerified(msg.sender);
86 }
87
88 function addStudent( infinite gas
89     string memory _firstName,
90     string memory _lastName,
91     string memory _email,
92     string memory _gender,
93     uint _absenceDays,
94     string memory _extracurricularActivities,
95     string memory _careerAspiration,

```

Fig 8.3: Implemented Code Snippet

```

96     uint[7] memory scores
97 ) public onlyTeacher {
98     studentCount++;
99     students[studentCount] = Student(
100         _firstName,
101         _lastName,
102         _email,
103         _gender,
104         _absenceDays,
105         _extracurricularActivities,
106         _careerAspiration,
107         scores[0],
108         scores[1],
109         scores[2],
110         scores[3],
111         scores[4],
112         scores[5],
113         scores[6],
114         scores[0] + scores[1] + scores[2] + scores[3] + scores[4] + scores[5] + scores[6],
115         (scores[0] + scores[1] + scores[2] + scores[3] + scores[4] + scores[5] + scores[6]) / 7
116     );
117     emit StudentAdded(studentCount);
118 }
119
120 function editStudent( infinite gas
121     uint studentId,
122     string memory _firstName,
123     string memory _lastName,
124     string memory _email,
125     string memory _gender,
126     uint _absenceDays,
127     string memory _extracurricularActivities,

```

Fig 8.4: Implemented Code Snippet


```
128     string memory _careerAspiration,
129     uint[7] memory scores
130 ) public onlyTeacher {
131     require(studentId <= studentCount, "Invalid student ID");
132
133     Student storage student = students[studentId];
134     student.firstName = _firstName;
135     student.lastName = _lastName;
136     student.email = _email;
137     student.gender = _gender;
138     student.absenceDays = _absenceDays;
139     student.extracurricularActivities = _extracurricularActivities;
140     student.careerAspiration = _careerAspiration;
141     student.mathScore = scores[0];
142     student.historyScore = scores[1];
143     student.physicsScore = scores[2];
144     student.chemistryScore = scores[3];
145     student.biologyScore = scores[4];
146     student.englishScore = scores[5];
147     student.geographyScore = scores[6];
148     student.totalScore = scores[0] + scores[1] + scores[2] + scores[3] + scores[4] + scores[5] + scores[6];
149     student.percentage = student.totalScore / 7;
150
151     emit StudentUpdated(studentId);
152 }
153
154 function getStudent(uint studentId) public view verifiedUser returns (Student memory) {  infinite gas
155     require(studentId <= studentCount, "Invalid student ID");
156     return students[studentId];
157 }
158 }
```

Fig 8.5: Implemented Code Snippet

GitHub Link:

Securing Student's Scores Using Blockchain <https://github.com/Aneesh997/Securing-Student-Scores-Using-Blockchain/tree/main>